What does heterogeneity cost? A controlled, communication-priced decomposition of federated LSTM load forecasting on London smart meters

Pushkal Gupta[a,*], Vanka Kaushik[a], Kalaavathi B[a]

[a] *Department of Internet of Things, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

Abstract

Federated learning is promoted for household load forecasting because it keeps smart-meter data at home. Yet its accuracy loss is usually blamed on data heterogeneity, and its bandwidth advantage is assumed rather than measured. We test both claims on 100 London households from the Low Carbon London trial with a controlled protocol: the same homes, test set, LSTM and compute budget, changing one factor at a time: number of clients, heterogeneity (dialled over real load-profile clusters and measured by a label-free index), participation, aggregation rule (FedAvg, FedProx, a heterogeneity-adaptive proximal rule) and update precision. Every byte exchanged is counted. Federated training on a single client matches centralised accuracy. The gap is caused by fragmentation, growing from 2 % with five clients to 10 % with one client per home. Heterogeneity at a fixed number of clients barely changes final error, but raises the traffic needed to approach centralised accuracy by about 85 %. Federated training moves more bytes than uploading the raw readings. FedAvg leaves most households worse off than local models; one epoch of local fine-tuning largely removes this at community scale. Proximal regularisation does not help, and its adaptive variant fails because update dissimilarity is dominated by convergence, not

*Corresponding author.

Email addresses: pushkal.gupta2023@vitstudent.ac.in (Pushkal Gupta), pushkalgupta2005@gmail.com (Pushkal Gupta), vanka.kaushik2023@vitstudent.ac.in (Vanka Kaushik), kalaavathi.b@vit.ac.in (Kalaavathi B)

heterogeneity.

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1. Introduction

Short-term load forecasts at the level of a single household feed demand response, home energy management, local flexibility markets and the operation of low-voltage networks that now host heat pumps, electric vehicles and rooftop photovoltaics. Deep recurrent models, in particular the long short-term memory (LSTM) network [1], forecast individual households better than per-home statistical models when they are trained on many households at once [2]. Pooling the readings in one place, however, is what smart-meter regulation and customer consent increasingly discourage: half-hourly consumption reveals occupancy, routines and appliance use. Federated learning (FL) [3, 4] offers an alternative. Each meter, home gateway or community energy hub trains on its own data, and only model weights travel to an aggregator.

A large body of work now applies FL to residential load forecasting [5, 6, 7, 8, 9, 10, 11, 12, 13, 14]. Two premises run through it. The first is that household data are strongly non-identically distributed (non-IID), so that plain federated averaging (FedAvg) suffers, and that remedies such as the proximal term of FedProx [15], clustering [7, 16] or personalisation [9, 8, 12] are needed. The second is that FL is the communication-light, privacy-preserving option. Both premises have been tested only in isolation. On LCL, Briggs et al. [17] found plain FedAvg worse than purely local models and recovered the loss with clustering and local fine-tuning. Taïk and Cherkaoui [18] and Savi and Olivadese [19] estimated large bandwidth savings over raw-data upload, whereas Briggs et al. [20] argued, without measurement, that weight vectors cost more bandwidth than half-hourly readings on constrained meter links. Studies differ in dataset, cohort, horizon, model, number of rounds and evaluation protocol, and almost none holds the households and the test set fixed while varying only how the data are split across clients. As a result, the field cannot say how much of the accuracy lost by FL is caused by heterogeneity, how much by splitting the data at all, and how much by the federated training protocol itself. Nor can it say whether the bytes exchanged during training are small compared with simply uploading

the raw readings.

This paper answers these questions with a controlled experiment on the Low Carbon London (LCL) smart-meter trial [21]. We fix a stratified cohort of 100 standard-tariff London households, one chronological train/validation/test split, one LSTM of 17,217 parameters and one training budget. We then change exactly one factor at a time: whether data are pooled, the number of clients the same homes are split into, how homes are grouped into clients (from random to fully clustered by load profile), client participation and stragglers, the aggregation rule, and the precision of the transmitted update. Every comparison uses the same test half-hours, so differences in error are attributable to the manipulated factor alone.

Contributions.

1. **A controlled decomposition protocol for federated load forecasting.** We separate the gap between federated and centralised training into the cost of the round protocol (one client holding all data), the cost of fragmentation (the same homes split over K clients), the cost of heterogeneity (a Dirichlet dial over real load-profile clusters) and the cost of partial participation. Heterogeneity is reported as a *measured* index H between clients, computed on training data only, rather than as a nominal partition label.
2. **Communication-priced accuracy.** Prior estimates of FL's bandwidth advantage are analytic, derived from model and data sizes [18], or summarised as an overall ratio of exchanged data [19]. We count every byte uploaded and downloaded and introduce two budget metrics, R@ε and B@ε: the rounds and megabytes a federation needs to come within ε of centralised validation error. We compare these with the volume of the raw training data itself.
3. **Equity of federation.** We report, household by household, who gains and who loses relative to training alone, by consumption quintile, with paired signed-rank tests [22] and Holm correction [23]. We also evaluate a zero-communication personalisation step, one epoch of local fine-tuning of the global model.
4. **An update-dissimilarity schedule for the proximal coefficient, and why it does not help here.** Adaptive proximal coefficients exist, driven by the training loss [15] or by a local-versus-global performance index [24]. HA-FedProx instead schedules μ from a parameter-free,

bounded, per-round estimate of FedProx's own B-dissimilarity [15], computed in update space from messages the server already receives. It is closely related to gradient diversity [25]. We show that on household load data this signal does rank heterogeneity but is dominated by training progress, and we explain why. This negative result applies to any scheme that adapts regularisation to observed update dissimilarity.

5. **An open, reproducible pipeline.** Preprocessing, partitioning, all arms, byte accounting, statistics, figures and the tables of this paper are generated by one public code base from the public LCL release. A Flower backend [26] reproduces the simulator exactly.

Main findings. Running the federated protocol on a single client that holds all homes matches centralised accuracy, so the round protocol costs nothing. The federated gap is caused by fragmentation: it grows with the logarithm of the number of clients, from about 2 % with five clients to about 10 % with one client per home. Heterogeneity between clients, dialled from random to fully clustered at a fixed number of clients, barely changes final accuracy but raises the traffic needed to approach centralised accuracy by about 85 %. At this model size, federated training moves more bytes than uploading the raw readings, so its case rests on data governance, not bandwidth. Plain FedAvg leaves most households worse off than training alone. One epoch of local fine-tuning, which needs no communication, largely removes that harm at community scale. Proximal regularisation brings no benefit, and its update-adaptive variant is worse. We show that the update-dissimilarity signal it relies on is dominated by convergence rather than by heterogeneity.

Section 2 reviews related work. Section 3 defines the task, the arms, the heterogeneity index, HA-FedProx and the communication metrics. Section 4 describes data and protocol. Section 5 reports the results, Section 6 discusses their implications for deployment, and Section 7 concludes.

2. Related work

2.1. Household load forecasting with deep learning

Household load is volatile and dominated by occupant behaviour. Kong et al. [2] showed that LSTMs outperform shallow and statistical baselines for individual residential customers. Shi et al. [27] showed that pooling the data of neighbouring households reduces over-fitting, the effect that motivates collaborative training in the first place. Later work added embeddings,

bidirectional recurrences and multi-task sharing [28, 29], as well as clustering of consumption profiles [30]. These gains assume the data are pooled.

2.2. Federated household load forecasting

FL for residential forecasting has grown quickly [5, 6, 7, 31, 10, 11, 9, 32, 33, 8, 16, 34, 35, 36, 37, 38, 39, 40, 41, 42, 12, 13, 14, 43, 44, 45, 46, 47, 48]. On the question that matters most here, whether federation pays off at all, these studies disagree. The closest to ours is that of Briggs et al. [17], who trained a two-layer LSTM with FedAvg on 100 LCL households. Plain FL was 7.6 % worse in mean RMSE than localised learning; clustering reduced the shortfall to 4.7 %, and local fine-tuning made the federated models better than the local ones. Gholizadeh and Musilek [7] also found local learning best on average for individual Canadian homes, while Fekri et al. [6] found FedAvg comparable to, or better than, per-meter models on 19 Ontario homes. Personalised and clustered variants [9, 8, 16, 12, 13, 44] are almost always motivated by heterogeneity among households. None of these studies varies heterogeneity while holding the households and the test set fixed, so the share of the federated gap caused by heterogeneity, rather than by data fragmentation or the training protocol, has not been measured. We revisit the LCL comparison of Briggs et al. under a controlled protocol, with per-household paired statistics, and add that decomposition.

2.3. Communication cost

Communication efficiency was a founding motivation of FL [3, 49]. In load forecasting the evidence on it points both ways. Taïk and Cherkaoui [18] estimated analytically that FL saves 90–97 % of the network load of uploading data, and Savi and Olivadese [19] reported up to 50 times less exchanged data at sub-hourly resolution. Briggs et al. [20], on the other hand, argued without measurement that exchanging weight vectors costs more bandwidth than sending half-hourly readings over constrained smart-meter links. Most FL forecasting studies count rounds rather than bytes [6], or report payload per round without comparing it with the data [50]. We measure upload and download bytes per round, the bytes needed to reach a given accuracy, and the raw-data volume they replace.

2.4. Non-IID data and proximal regularisation

Statistical heterogeneity slows and biases FedAvg [51, 52, 4]. The remedies proposed include proximal regularisation (FedProx [15]), control variates

(SCAFFOLD [53]), dynamic regularisation (FedDyn [54]), normalised averaging (FedNova [55]), adaptive server optimisers [56], personalisation [57, 58] and fairness-aware objectives [59]. Heterogeneity is usually induced with a Dirichlet split over class labels [60]. Regression on smart-meter data has no class labels, so we use a Dirichlet split over load-profile clusters and also measure the resulting heterogeneity directly.

Choosing the proximal coefficient remains ad hoc. The FedProx paper adapts μ from the training loss [15]. HAPI-FedProx adapts the proximal term from a local-versus-global performance index [24]. Yu et al. [61] characterise the optimal regularisation strength as a function of heterogeneity in theory. FedAdp [62] uses the geometry of client updates to set aggregation weights. The update-disagreement ratio used by HA-FedProx is a bounded, update-space estimate of FedProx's B-dissimilarity, and it belongs to the same family as gradient diversity [25]. To our knowledge it has not been used to schedule μ, and its behaviour on real energy data has not been examined.

2.5. Privacy

FL keeps raw data local, but updates can leak training data through gradient inversion [63, 64]. We therefore claim data locality only, not formal privacy.

3. Methodology

3.1. Forecasting task and model

Let $x_{i,t}$ be the energy (kWh) consumed by household i in half-hour t. Each household's series is standardised with a mean and standard deviation estimated on its own training period only. The forecaster receives the last $L = 48$ standardised readings (24 h) and predicts the next half-hour, $\hat{x}_{i,t+1} = f_w(x_{i,t-L+1}, \ldots, x_{i,t})$. All arms share one architecture: a single-layer LSTM with 64 hidden units followed by a linear output layer ($P = 17{,}217$ parameters, 68.9 kB in 32-bit floating point). Training minimises the mean squared error with Adam (learning rate 10^{-3}, batch 256). Errors are reported in kWh after inverting each household's scaling. Keeping the model small and fixed is deliberate: the object of study is the training regime, not the architecture.

3.2. Arms

Naive baselines. Persistence ($\hat{x}_{t+1} = x_t$) and seasonal naive ($\hat{x}_{t+1} = x_{t-47}$).

Local-only. One model per household, trained on that household's windows only, with early stopping on its own validation error. No data or weights leave the home.

Centralised. One model trained on the pooled windows of all households, with early stopping on pooled validation error. This is the accuracy reference, and it requires the raw readings to leave the home.

FedAvg. K clients each hold the windows of one or more households. In round t the server samples a set S_t of clients, sends them the global weights w_t, each client trains for E local epochs and returns w_k, and the server sets $w_{t+1} = \sum_{k \in S'_t} p_k w_k$ with $p_k = n_k / \sum_{j \in S'_t} n_j$, where n_k is the number of training windows of client k and $S'_t \subseteq S_t$ the clients that returned [3].

FedProx. Each client minimises $F_k(w) + \frac{\mu}{2}\|w - w_t\|^2$ and stragglers' partial updates are kept rather than dropped [15].

HA-FedProx. FedProx with a coefficient μ_t set by the server each round (Section 3.5).

FedAvg + local fine-tuning (FT). After federated training, each household fine-tunes the final global model for one epoch on its own training windows. This costs no communication and is evaluated for every federated run.

Clients are stateless: a fresh optimiser is created at every local update, as is standard for cross-device FL.

3.3. Clients and controlled partitions

Two client models are considered. In *household mode* each home is a client ($K = 100$), which reflects meter-level or home-gateway deployment and has natural, uncontrolled heterogeneity. In *community mode* the same homes are grouped into K clients, which reflects a building gateway, a community energy hub or an aggregator.

To dial heterogeneity with real, not invented, labels, every household is described by its normalised 24-bin mean daily load shape and its log mean

load, both computed on the training period. These features are standardised and clustered with k-means ($k = 5$, 10 restarts) into load-profile groups such as evening-peaking or flat, low or high consumption. Households are then assigned to clients by one of three strategies: (i) *random*, where homes are shuffled into equal groups; (ii) *Dirichlet*(α) [60], where each client draws mixture proportions $\pi_k \sim \mathrm{Dir}(\alpha \mathbf{1})$ over profile clusters and fills its slots by sampling a cluster from π_k, renormalised to clusters with homes left; and (iii) *cluster*, where homes are sorted by cluster and mean load and chunked, the maximally separated case. Because the homes, and therefore the test windows, are identical under every strategy, error differences between partitions are caused by the partition alone.

3.4. Measured heterogeneity index

Nominal labels such as "$\alpha = 0.3$" do not say how different the resulting clients are. We therefore measure heterogeneity directly on training data. For client k, let $q_k \in \Delta^{167}$ be the distribution of its total training consumption over the 168 hours of the week. With JS the Jensen–Shannon divergence in base 2 [65] and m_k the mean load of client k,

$$H_{\mathrm{JS}} = \binom{K}{2}^{-1} \sum_{k<l} \mathrm{JS}(q_k \,\|\, q_l), \qquad H_{\mathrm{CV}} = \frac{\mathrm{sd}(m_k)}{\mathrm{mean}(m_k)}, \tag{1}$$

$$H = H_{\mathrm{JS}} + \frac{H_{\mathrm{CV}}}{1 + H_{\mathrm{CV}}}. \tag{2}$$

$H_{\mathrm{JS}} \in [0, 1]$ measures differences in *when* clients consume, and the bounded transform of H_{CV} measures differences in *how much* they consume. $H = 0$ for a single client or identical clients. The index uses no labels and no test data.

3.5. HA-FedProx: a heterogeneity-adaptive proximal coefficient

The proximal coefficient of FedProx trades client drift against local progress. It is usually fixed by tuning, yet the appropriate strength should depend on how much the client updates disagree. After aggregating round t, the server holds the client updates $\Delta_k = w_k - w_t$ and their weighted mean $\bar{\Delta} = \sum_k p_k \Delta_k$. It computes the *update-disagreement ratio*

$$\rho_t = \frac{\sum_k p_k \|\Delta_k - \bar{\Delta}\|^2}{\sum_k p_k \|\Delta_k\|^2} \in [0, 1], \tag{3}$$

which is 0 when all clients move identically and approaches 1 when their updates cancel. By the bias–variance identity $\sum_k p_k \|\Delta_k\|^2 = \|\bar{\Delta}\|^2 + \sum_k p_k \|\Delta_k - \bar{\Delta}\|^2$, $\rho_t = 1 - 1/B_t^2$ with $B_t^2 = \sum_k p_k \|\Delta_k\|^2 / \|\bar{\Delta}\|^2$, an update-space analogue of the B-local dissimilarity used in the convergence analysis of FedProx [15]. The next round uses

$$\mu_{t+1} = \mathrm{clip}(\mu_0 \, \rho_t, \ \mu_{\min}, \ \mu_{\max}) \,, \tag{4}$$

with $\mu_{\min} = 10^{-4}$, $\mu_{\max} = 1$ and μ_0 tuned on validation data exactly as FedProx's fixed μ. The rule needs no additional client computation or messages, and it reduces to FedAvg for $\mu_0 = 0$. An earlier variant normalised by $\|\bar{\Delta}\|^2$ alone; it is unbounded as the global model converges and $\|\bar{\Delta}\| \to 0$, which is why the bounded ratio (3) is used.

3.6. Partial participation and stragglers

In each round a fraction C of clients is sampled. Each sampled client independently becomes a straggler with probability p_s and completes only a uniform random fraction $u \sim U(0.1, 1)$ of its local work. Following the protocol of Li et al. [15], FedAvg drops stragglers (they download but never upload) while FedProx and HA-FedProx aggregate their partial updates.

3.7. Communication accounting

A model message carries $P \cdot b$ bytes, with $b = 4$ for 32-bit and $b = 2$ for 16-bit updates. Round t costs $Pb(|S_t| + |S'_t|)$: one download per selected client and one upload per returning client. Protocol framing is excluded; the Flower backend measures it at about 1 % of the payload. For the centralised arm we report the volume of the raw training readings, at 4 bytes per value plus an 8-byte timestamp, as the bytes that pooling would move. Let v_t be the validation MAE of the global model after round t and $v^\star$ the centralised validation MAE. We define

$$R@\varepsilon = \min\{t : v_t \le (1+\varepsilon) v^\star\}, \qquad B@\varepsilon = \sum_{s \le R@\varepsilon} Pb(|S_s| + |S'_s|), \tag{5}$$

with $\varepsilon = 10\,\%$. These convert "how close to centralised" into "at what network price".

3.8. Metrics and statistics

We report mean absolute error (MAE), root mean squared error (RMSE), weighted absolute percentage error (WAPE $= 100 \sum |x - \hat{x}| / \sum |x|$), and MAPE restricted to half-hours with $x \geq 0.05\,\text{kWh}$. Unrestricted MAPE is unbounded at the near-zero night-time loads that are common in households. All pooled metrics are micro-averaged over every test half-hour of every household. Per-household errors are averaged over seeds and compared with two-sided Wilcoxon signed-rank tests [22] over the 100 households, with Holm correction [23] across the family of comparisons. Because p-values with 100 paired households can be small for negligible effects, every test is reported with its median paired difference and the share of households for which one arm is better.

4. Experimental setup

4.1. Data

We use the public release of the Low Carbon London smart-meter trial run by UK Power Networks [21]. It contains half-hourly consumption of 5,567 London households between November 2011 and February 2014. We stream all 168 files of the partitioned release (93.1 million readings in the study year), observe 5,528 household identifiers, and apply the following rules. The study period is calendar year 2013. Only standard-tariff households are kept, because the time-of-use group received dynamic price signals in 2013 that would confound the comparison. Unparseable, negative or implausible readings ($> 10\,\text{kWh}$ per half-hour) are set to missing, duplicate timestamps are averaged, and each series is re-indexed to a full 30-minute grid. Gaps of up to 2 h are linearly interpolated and longer gaps are left missing; any window that touches a missing value is dropped. Households with at least 95 % coverage in 2013 are eligible (4,034 homes). From these we draw a cohort of 100 homes, stratified into quintiles of training-period mean load (20 per quintile) with a fixed selection seed. The cohort's mean training-period load is 0.206 kWh per half-hour (range 0.008–0.972), and its mean coverage is 99.9 %.

The split is chronological and identical for every household: training January–August 2013, validation September–October and test November–December. A window belongs to the split of its target time stamp, so its inputs may reach back into the previous split; those readings are known at forecast time. This yields 1,160,952 training and 292,052 test windows.

Table 1: Experimental matrix. All rows use the same 100 households, split, model and budget.

ID	Arm(s)	Factor varied
E0	persistence, seasonal naive	reference
E1	local-only	no collaboration
E2	centralised	pooled raw data
E3	FedAvg	random grouping, $K = 20$
E4, E5	FedAvg, FedProx	Dirichlet $\alpha = 0.3$
E6, E7	FedAvg, FedProx	cluster grouping (strong non-IID)
E8, E9, E11	FedAvg, FedProx, HA-FedProx	$\alpha = 0.3$, $C = 0.5$, 20 % stragglers
E10	FedAvg, FedProx, HA-FedProx	$\alpha \in \{\infty, 1, 0.3, 0.1, 0\}$ sweep
E12	FedAvg	$\alpha = 0.3$, 16-bit updates
E13	FedAvg, FedProx, HA-FedProx	one household per client ($K = 100$)
E14	FedAvg	random grouping, $K \in \{1, 5, 50\}$

Scalers, load-profile clusters and the heterogeneity index are computed from training data only, and every decision (early stopping, best federated round, hyper-parameters) uses validation error only. Test error never drives a decision.

4.2. Experimental matrix

Table 1 lists the experiments. Unless stated otherwise, community mode uses $K = 20$ clients of five homes each, every client participates in every round, $E = 1$ local epoch and $T = 30$ rounds. Thirty rounds of one local epoch equal the number of passes over the data allowed to the centralised arm (30 epochs, early stopping with patience 4), so federated and centralised training have the same computational budget. FedProx's μ and HA-FedProx's μ_0 were tuned on validation data from the grids $\{10^{-3}, 10^{-2}, 10^{-1}\}$ and $\{10^{-2}, 10^{-1}, 1\}$, under the hardest (cluster) partition and with the first seed. This selected $\mu = 0.001$ and $\mu_0 = 0.01$, which were then used in every other experiment. Each experiment is repeated with three seeds (42, 43, 44), which change model initialisation, client sampling, stragglers and the random and Dirichlet groupings, but not the cohort, the profile clusters or the split.

4.3. Implementation

The pipeline is written in Python with PyTorch. The federated loop is a deterministic in-process simulator whose clients implement the Flower

`NumPyClient` interface. An optional backend runs the same clients through Flower's own FedAvg and FedProx strategies [26], and on a test configuration it reproduces the simulator's per-round validation error and final test error exactly. Experiments ran on a single Apple M5 Max workstation (Metal backend), eight runs in parallel. Code, configuration files and every result table are available in the accompanying repository (Section 7).

5. Results

All numbers are means over three seeds on the 292,052 test half-hours of the 100 households. The standard deviations over seeds are given in the tables. Gaps are expressed relative to the centralised test MAE.

5.1. Accuracy of the arms

Table 2 compares all arms. Every LSTM beats both naive references. Persistence (0.0890 kWh) is a much stronger baseline than seasonal naive (0.1332 kWh) at a 30-minute horizon, because consecutive half-hours are strongly correlated. The centralised LSTM is the most accurate arm (MAE 0.0811 kWh). It is better than local-only training (0.0840 kWh) for 72 % of households (Holm-adjusted $p < 10^{-4}$), which confirms that cross-household data are valuable for this task.

Federated training recovers only part of that value. FedAvg with random grouping reaches 0.0853 kWh, +5.2 % above centralised, and it is worse than centralised for every household. Under moderate heterogeneity ($\alpha = 0.3$) FedAvg reaches 0.0861 kWh and is worse than local-only training on pooled MAE. Per household it beats local-only for only 34 % of homes (p_{Holm} 0.0023). On RMSE the ordering reverses: FedAvg (0.1758 kWh) beats local-only (0.1779 kWh). The shared model handles large deviations better, while per-home models fit the typical half-hour better. Our protocol therefore reproduces the observation of Briggs et al. [17] that plain FL can lose to local models on LCL, but only for MAE, the absolute-error criterion.

Fig. 1 shows the first three test days of the household whose mean load is closest to the cohort median. All learned models track the daily pattern and lag sharp, short spikes by one step, which is typical of one-step forecasters of household load. The differences between arms are small relative to the volatility of the series, which is why the paired, per-household tests below are needed to separate them.

Table 2: Test error of all arms (kWh per half-hour; mean ± s.d. over three seeds). MAPE* is computed over half-hours with load ≥ 0.05 kWh. Traffic is the cumulative upload plus download over all rounds; for the centralised arm it is the raw training-data volume.

Method	MAE	RMSE	MAPE*	WAPE	Best rd.	Traffic (MB)
Persistence (naive)	0.0890	0.1987	41.1	37.6	n/a	0.0
Seasonal naive (t–48)	0.1332	0.2773	63.6	56.3	n/a	0.0
Local-only	0.0840 ± 0.0003	0.1779 ± 0.0006	37.3 ± 0.4	35.5	n/a	0.0
Centralised	0.0811 ± 0.0003	0.1718 ± 0.0015	36.7 ± 0.8	34.3	n/a	14.0
FedAvg, random	0.0853 ± 0.0009	0.1758 ± 0.0007	39.0 ± 0.6	36.1	23.0 / 30	82.6
FedAvg, α=0.3	0.0861 ± 0.0004	0.1758 ± 0.0007	39.7 ± 0.1	36.4	24.0 / 30	82.6
FedProx, α=0.3	0.0858 ± 0.0001	0.1755 ± 0.0006	39.5 ± 0.1	36.3	26.0 / 30	82.6
HA-FedProx, α=0.3	0.0863 ± 0.0003	0.1765 ± 0.0005	39.6 ± 0.1	36.5	20.7 / 30	82.6
FedAvg, cluster	0.0857 ± 0.0003	0.1758 ± 0.0005	39.2 ± 0.2	36.2	22.0 / 30	82.6
FedProx, cluster	0.0858 ± 0.0002	0.1759 ± 0.0006	39.3 ± 0.2	36.3	23.3 / 30	82.6
HA-FedProx, cluster	0.0865 ± 0.0002	0.1772 ± 0.0008	39.5 ± 0.4	36.6	15.7 / 30	82.6
FedAvg, α=0.3, partial	0.0849 ± 0.0004	0.1757 ± 0.0010	38.7 ± 0.5	35.9	24.0 / 30	37.4
FedProx, α=0.3, partial	0.0857 ± 0.0003	0.1764 ± 0.0012	39.1 ± 0.3	36.2	20.7 / 30	41.3
HA-FedProx, α=0.3, partial	0.0857 ± 0.0003	0.1769 ± 0.0011	39.1 ± 0.2	36.2	23.0 / 30	41.3
FedAvg, α=0.3, fp16	0.0858 ± 0.0005	0.1756 ± 0.0009	39.5 ± 0.3	36.3	25.3 / 30	41.3
FedAvg, K=100	0.0894 ± 0.0013	0.1816 ± 0.0009	40.9 ± 1.5	37.8	29.0 / 30	413.2
FedProx, K=100	0.0893 ± 0.0013	0.1817 ± 0.0011	40.7 ± 1.6	37.8	30.0 / 30	413.2
HA-FedProx, K=100	0.0901 ± 0.0005	0.1823 ± 0.0009	41.2 ± 0.8	38.1	29.3 / 30	413.2
FedAvg, K=1, random	0.0808 ± 0.0005	0.1710 ± 0.0003	36.7 ± 0.3	34.2	16.3 / 30	4.1
FedAvg, K=5, random	0.0827 ± 0.0005	0.1721 ± 0.0006	38.1 ± 0.2	35.0	23.0 / 30	20.7
FedAvg, K=50, random	0.0869 ± 0.0006	0.1782 ± 0.0005	39.6 ± 0.5	36.8	26.7 / 30	206.6

5.2. Where the federated gap comes from

Table 4 walks from centralised training to the hardest federated setting, changing one factor at a time. Fig. 2 shows the fragmentation sweep.

Running the federated protocol with a single client that holds all 100 homes ($K = 1$) reproduces centralised accuracy: 0.0808 kWh against 0.0811 kWh, with no significant per-household difference (52 % of homes better, p_{Holm} 0.3138). Rounds, a fresh optimiser at every round and best-round selection therefore cost nothing. The gap appears as soon as the same data are split. It grows steadily with the number of clients, from +2.0 % at $K = 5$ through +5.2 % at $K = 20$ and +7.2 % at $K = 50$ to +10.3 % when every household is its own client (Table 3, Fig. 2). At $K = 20$, FedAvg with random grouping is worse than the single-client run for 99 % of households.

Splitting homes over more clients also raises the measured heterogeneity between clients, so the two effects must be separated. A regression of per-run FedAvg test MAE on $\log K$ and H_{JS} over all full-participation runs ($n = 24$, $R^2 = 0.88$) attributes the error to fragmentation: each unit of $\log K$ adds 1.33×10^{-3} kWh (p 0.0070), while the heterogeneity term is not significant (p

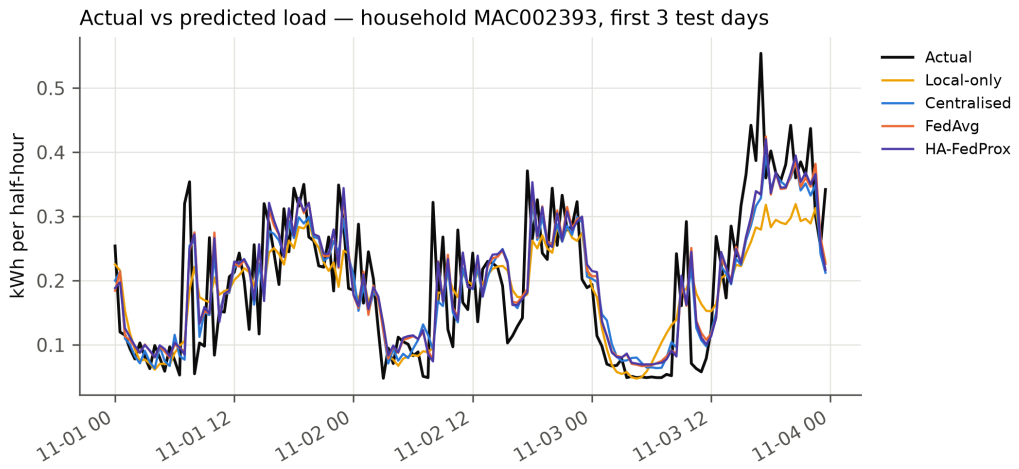


Figure 1: Actual and predicted load for the household closest to the cohort's median load (chosen without reference to any error), first three test days.

0.0553). Within $K = 20$, where heterogeneity was dialled independently of K, error and H_{JS} are unrelated (Spearman +0.23, p 0.4108, $n = 15$). A direct comparison makes the same point: 50 clients of two randomly grouped homes ($H = 0.354$) are worse than 20 clients grouped by load profile ($H = 0.381$), 0.0869 against 0.0857 kWh.

We attribute the effect to the averaging step. With one local epoch per round, each client takes about $n_k/256$ optimiser steps on its own data before its model is averaged with the others. The smaller the client, the fewer and noisier its steps, and the further the average of K such models falls from the model that a single pass over the pooled data would produce. The total amount of data and computation is the same for every K; what changes is how much of it is combined by averaging weights rather than by following gradients.

5.3. Heterogeneity changes the price of accuracy, not accuracy

Table 5 and Fig. 3 report the heterogeneity sweep. The load-shape divergence between clients, H_{JS}, increases monotonically from random grouping to the cluster partition, by a factor of about two. The composite index H does not, because random grouping produces a larger spread of client mean load than the Dirichlet groupings, and the cluster partition, which sorts homes by load level, produces the largest spread. Nominal labels such as α are there-

Table 3: Fragmentation sweep: the same 100 households split over K clients with random grouping. $K = 100$ is household mode. FT: one epoch of local fine-tuning of the final global model.

K	homes/client	H_{JS}	Test MAE	FT MAE	$\bar{\rho}$	Gap to central (%)
1	100	0.0000	0.0808 ± 0.0005	0.0812	0.000	-0.3
5	20	0.0028	0.0827 ± 0.0005	0.0818	0.759	+2.0
20	5	0.0105	0.0853 ± 0.0009	0.0834	0.856	+5.2
50	2	0.0261	0.0869 ± 0.0006	0.0842	0.884	+7.2
100	1	0.0497	0.0894 ± 0.0013	0.0851	0.890	+10.3

Table 4: Decomposition of the federated–centralised gap (FedAvg, test MAE in kWh). Each row adds one factor to the row above. Rows without a Δ are alternatives to the row above, not additions.

Setting	Test MAE	Δ vs previous	Gap to central (%)
Centralised (pooled raw data)	0.0811 ± 0.0003	–	+0.0
+ round protocol, 1 client (K=1)	0.0808 ± 0.0005	-0.0002	-0.3
+ split over 20 clients, random grouping	0.0853 ± 0.0009	+0.0044	+5.2
+ label-skewed grouping (α=0.3)	0.0861 ± 0.0004	+0.0008	+6.2
+ 50% participation, 20% stragglers	0.0849 ± 0.0004	-0.0012	+4.7
cluster grouping instead (strong non-IID)	0.0857 ± 0.0003	–	+5.7
1 home = 1 client (K=100)	0.0894 ± 0.0013	–	+10.3

fore an unreliable proxy for how different clients actually are, which is why we report measured values.

Across this whole range, the final FedAvg error moves by at most 0.0012 kWh (from 0.0849 to 0.0861 kWh), comparable to the seed-to-seed standard deviation, and does not trend with H_{JS} (Fig. 3a). Per household, FedAvg under the cluster partition is better than under random grouping for 45 % of homes, and the difference is not significant after correction (p_{Holm} 0.0884). Heterogeneity does, however, change how fast the federation gets there. The traffic needed to come within 10 % of centralised validation MAE grows from 12.9 MB (4.7 rounds) with random grouping to 21.1 MB at $\alpha = 0.3$ and 23.9 MB under the cluster partition (Fig. 3b), an increase of about 85 %. In this setting, heterogeneity is a communication cost rather than an accuracy cost.

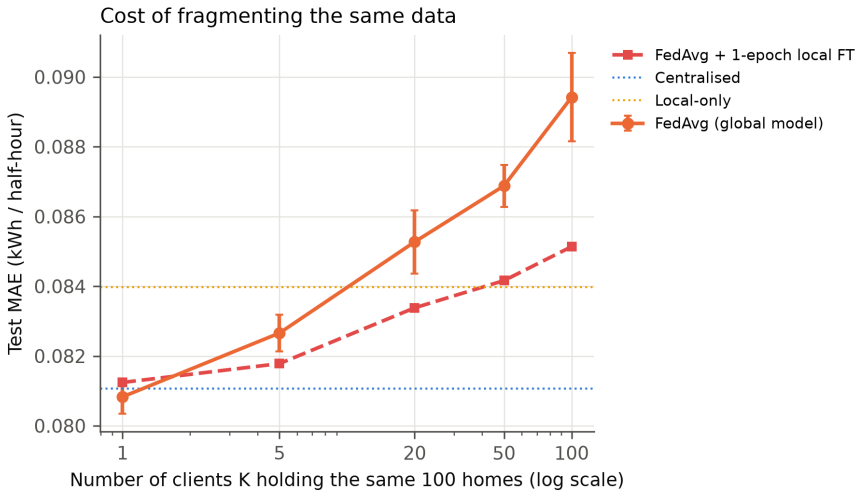


Figure 2: Test MAE when the same 100 households are split over K clients with random grouping ($K = 100$ is one household per client). The dashed line adds one epoch of local fine-tuning to the final global model.

5.4. Proximal regularisation and HA-FedProx

Validation-based tuning selected the smallest proximal coefficient in the grid for both methods ($\mu = 0.001$, $\mu_0 = 0.01$). Larger values consistently increased validation error, so the data asked for as little proximal pull as possible. With the tuned coefficient, FedProx and FedAvg are practically indistinguishable (Table 6). Under $\alpha = 0.3$ FedProx is better for 75 % of households, but the median paired difference is -0.0001 kWh, about 0.1 % of the error. Under the cluster partition and under partial participation, FedProx is slightly worse.

HA-FedProx never improves on FedAvg. Under the cluster partition it is worse for 73 % of households (median +0.0004 kWh, $p_{\text{Holm}} < 10^{-4}$), and under mild heterogeneity it needs more traffic to reach the ε band (35.8 MB with random grouping against 12.9 MB for FedAvg; Fig. 3b). Fig. 4 shows why. The update-disagreement ratio ρ_t starts low (0.15 on average in round 1), rises quickly and settles near 0.96 as the global model converges and the common descent direction $\bar{\Delta}$ shrinks relative to the client-specific parts of the updates. The signal does rank heterogeneity correctly: across the full-participation runs with $K = 20$, mean ρ increases with H_{JS} (Spearman +0.77, p 0.0002, $n = 18$), and the round-1 value even more so (Spearman +0.88).

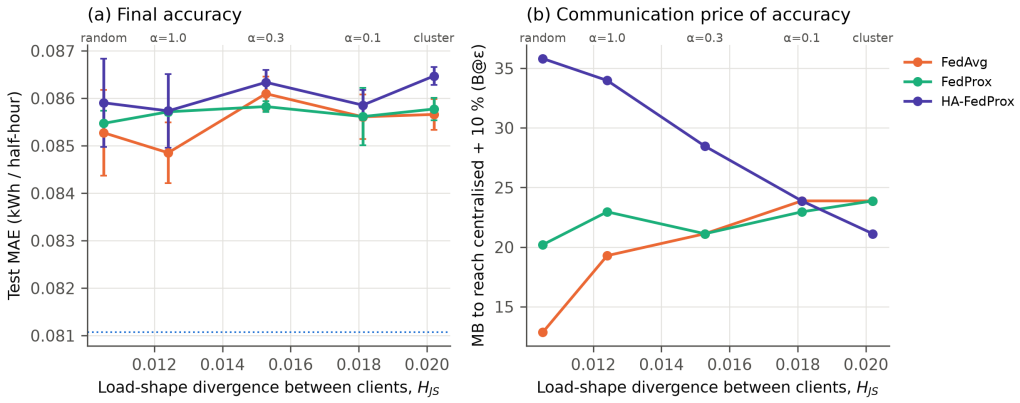


Figure 3: Heterogeneity sweep (community mode, $K = 20$). (a) Final test MAE against the measured load-shape divergence between clients; the dotted line is centralised training. (b) Megabytes exchanged before validation MAE first comes within 10 % of centralised (B@ε). Labels give the grouping strategy.

However, the spread of mean ρ across groupings is only 0.020, whereas ρ_t rises by 0.81 within a typical run, about 40 times more. A rule $\mu_{t+1} \propto \rho_t$ is therefore, in effect, a schedule that increases the proximal pull as training converges, whatever the heterogeneity. Late in training this suppresses useful local progress, which is the opposite of what the validation-tuned FedProx results ask for.

5.5. Communication

One model message is 68.9 kB. Thirty rounds of full participation with $K = 20$ clients therefore move 82.6 MB, about six times the 14.0 MB of raw training readings that centralised training would upload once. The bytes needed for near-centralised accuracy are much smaller, and comparable to the data volume: 12.9 MB with random grouping and 23.9 MB under strong heterogeneity (Fig. 5). Partial participation halves the traffic per round. FedAvg with 50 % participation and 20 % stragglers reaches the ε band after 6.2 MB and ends at 0.0849 kWh, no worse than full participation, using 37.4 MB in total. Sending updates as 16-bit floats halves the traffic (41.3 MB in total, 11.5 MB to the ε band) and does not harm accuracy: 0.0858 kWh against 0.0861 kWh at 32 bits. The 16-bit run is even marginally better for 92 % of households (median -0.0002 kWh), possibly because rounding acts as a mild regulariser.

Table 5: Heterogeneity sweep: measured indices, mean update disagreement $\bar{\rho}$, test MAE and gap to centralised, for each grouping and algorithm.

Partition	Method	H	H_{JS}	$\bar{\rho}$	Test MAE	Gap to central (%)
α=0.1	FedAvg	0.242	0.0181	0.876	0.0856 ± 0.0005	+5.6
α=0.1	FedProx	0.242	0.0181	0.863	0.0856 ± 0.0006	+5.6
α=0.1	HA-FedProx	0.242	0.0181	0.844	0.0859 ± 0.0003	+5.9
α=0.3	FedAvg	0.243	0.0153	0.874	0.0861 ± 0.0004	+6.2
α=0.3	FedProx	0.243	0.0153	0.861	0.0858 ± 0.0001	+5.9
α=0.3	HA-FedProx	0.243	0.0153	0.844	0.0863 ± 0.0003	+6.5
random	FedAvg	0.255	0.0105	0.856	0.0853 ± 0.0009	+5.2
random	FedProx	0.255	0.0105	0.845	0.0855 ± 0.0003	+5.4
random	HA-FedProx	0.255	0.0105	0.822	0.0859 ± 0.0009	+6.0
α=1.0	FedAvg	0.270	0.0124	0.867	0.0849 ± 0.0006	+4.7
α=1.0	FedProx	0.270	0.0124	0.856	0.0857 ± 0.0001	+5.7
α=1.0	HA-FedProx	0.270	0.0124	0.841	0.0857 ± 0.0008	+5.8
cluster	FedAvg	0.381	0.0202	0.876	0.0857 ± 0.0003	+5.7
cluster	FedProx	0.381	0.0202	0.866	0.0858 ± 0.0002	+5.8
cluster	HA-FedProx	0.381	0.0202	0.851	0.0865 ± 0.0002	+6.7

Fragmentation dominates the bill. One client moves only 4.1 MB over 30 rounds. With household clients the federation moves 413.2 MB, about 30 times the raw training data. Only two of the three household-mode FedAvg runs reached the ε band within 30 rounds; those took 28.5 rounds and 392.5 MB on average. None of the HA-FedProx household runs reached it.

Validation error does not decrease monotonically with traffic. All federated variants pass through a transient rise in rounds 5–8 before converging (Fig. 5). This transient coincides with the dip in ρ_t (Fig. 4), when client updates briefly align. It means that the round budget, and hence the traffic, must be chosen on validation error rather than fixed in advance.

5.6. Who gains from federation?

Table 7 and Fig. 6 report the per-household view. Centralised training leaves 28 % of households worse off than training alone. FedAvg leaves 56 % (random grouping) to 66 % ($\alpha = 0.3$) worse off. A single epoch of local fine-tuning of the final global model, which needs no communication, reduces this to 46–48 %. After fine-tuning, FedAvg is statistically indistinguishable from local training per household (52 % better, p_{Holm} 0.4960) and improves pooled MAE to 0.0834 kWh. It remains behind centralised training (27 % of households better). With household clients the picture is worse. FedAvg

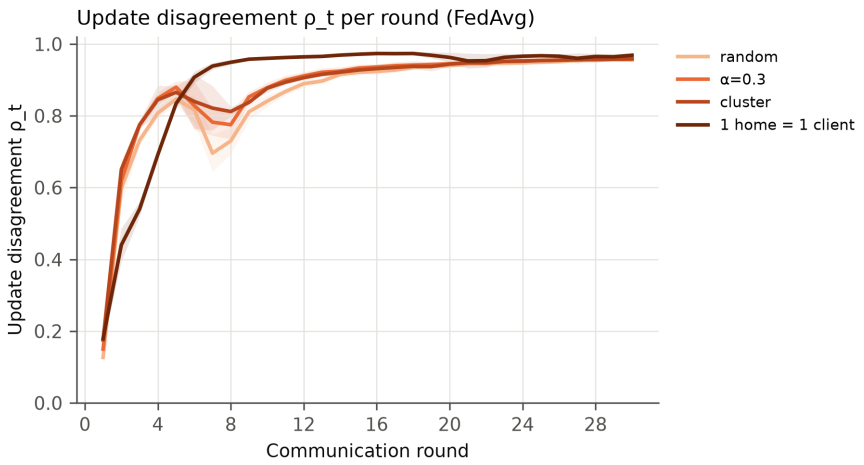


Figure 4: Update-disagreement ratio ρ_t per round for FedAvg under different groupings (mean ± s.d. over seeds). The within-run rise dominates the differences between groupings.

is worse than local training for 82 % of households ($p_{\text{Holm}} < 10^{-4}$), and its 90th-percentile household error rises to 0.1513 kWh, against 0.1441 kWh for local training. Fine-tuning reduces the share harmed to 61 % and the pooled MAE to 0.0851 kWh, which is still not better than local training (39 % of households better, p_{Holm} 0.0854).

By consumption quintile (Fig. 6b), the advantage of pooling the raw data is concentrated in the highest-consuming homes, whose errors are largest in absolute terms. FedAvg does not reproduce this advantage: in the top quintile its median household error is above that of local training. The households that federation fails most are therefore the ones whose forecasts matter most for network operation.

6. Discussion

6.1. What heterogeneity costs, and what it does not

The usual explanation for FedAvg's shortfall in household forecasting is that homes are too different. Our controlled experiment does not support that explanation for final accuracy. Holding everything else fixed, doubling the load-shape divergence between clients changed FedAvg's test error by at most 0.0012 kWh, comparable to the seed-to-seed variation. The fragmentation sweep shows where the gap does come from. The federated protocol with

Table 6: Paired Wilcoxon signed-rank tests over the 100 households (per-household test MAE averaged over seeds). "A better" is the share of households for which A has the lower MAE. p_{Holm} is Holm-adjusted across all rows.

A	B	median ΔMAE (A−B)	A better (%)	p	p_{Holm}
FedAvg, α=0.3	Local-only	+0.0013	34	0.0002	0.0023
FedAvg, α=0.3	Centralised	+0.0026	0	$< 10^{-4}$	$< 10^{-4}$
FedAvg, random	Centralised	+0.0021	0	$< 10^{-4}$	$< 10^{-4}$
FedProx, α=0.3	FedAvg, α=0.3	-0.0001	75	$< 10^{-4}$	$< 10^{-4}$
FedProx, cluster	FedAvg, cluster	+0.0001	36	0.0011	0.0103
FedProx, α=0.3, partial	FedAvg, α=0.3, partial	+0.0003	24	$< 10^{-4}$	$< 10^{-4}$
HA-FedProx, α=0.3	FedAvg, α=0.3	+0.0000	44	0.2480	0.4960
HA-FedProx, α=0.3	FedProx, α=0.3	+0.0001	43	0.0031	0.0217
HA-FedProx, cluster	FedAvg, cluster	+0.0004	27	$< 10^{-4}$	$< 10^{-4}$
HA-FedProx, cluster	FedProx, cluster	+0.0001	39	0.0012	0.0103
HA-FedProx, α=0.3, partial	FedAvg, α=0.3, partial	+0.0003	21	$< 10^{-4}$	$< 10^{-4}$
HA-FedProx, α=0.3, partial	FedProx, α=0.3, partial	+0.0002	35	0.0469	0.1875
FedAvg, cluster	FedAvg, random	+0.0001	45	0.0177	0.0884
FedAvg, K=100	Local-only	+0.0027	18	$< 10^{-4}$	$< 10^{-4}$
FedAvg, K=100	Centralised	+0.0045	0	$< 10^{-4}$	$< 10^{-4}$
HA-FedProx, K=100	FedAvg, K=100	+0.0003	22	$< 10^{-4}$	$< 10^{-4}$
FedAvg, α=0.3, fp16	FedAvg, α=0.3	-0.0002	92	$< 10^{-4}$	$< 10^{-4}$
Centralised	Local-only	-0.0016	72	$< 10^{-4}$	$< 10^{-4}$
FedAvg, K=1, random	Centralised	-0.0000	52	0.1046	0.3138
FedAvg, random	FedAvg, K=1, random	+0.0023	1	$< 10^{-4}$	$< 10^{-4}$
FedAvg, α=0.3 + local FT	Local-only	-0.0004	52	0.3900	0.4960
FedAvg, α=0.3 + local FT	Centralised	+0.0010	27	$< 10^{-4}$	$< 10^{-4}$
FedAvg, K=100 + local FT	Local-only	+0.0006	39	0.0142	0.0854

one client matches centralised training, and the error then grows with the logarithm of the number of clients the same data are spread over, whether or not those clients differ. Heterogeneity did matter for the speed of convergence: the traffic needed to approach centralised accuracy grew by about 85 % from random to clustered grouping. In practice this means that heterogeneity should be budgeted as a communication and latency cost. It is not a reason to expect a worse final model, provided that the round budget is large enough and the best round is chosen on validation data.

This also explains why remedies aimed at client drift showed little benefit here. FedProx helped only when its coefficient was close to zero, and the adaptive variant performed worse. When fragmentation, rather than drift between heterogeneous clients, causes most of the gap, a drift penalty has little to correct and mainly slows local progress.

6.2. Implications for deployment

The most consequential design choice in our experiments was the number of clients the data were spread over. Aggregating homes behind a building gateway, a community energy hub or a supplier-side aggregator, and federating those ($K = 5$ here), cost +2.0 % in MAE relative to centralised training

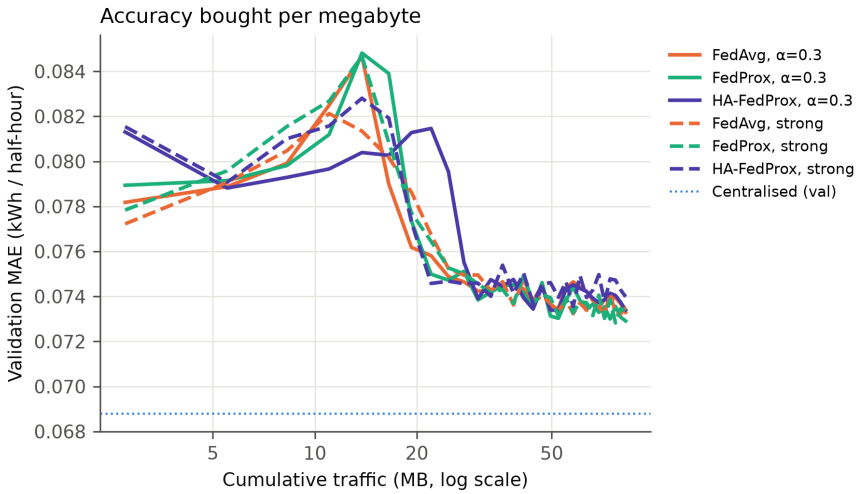


Figure 5: Validation MAE of the global model against cumulative traffic (upload plus download, log scale). The dotted line is the centralised validation MAE.

and needed a quarter of the traffic of 20 clients (20.7 MB). Meter-level federation ($K = 100$) lost +10.3 % and moved about 30 times the raw data. Where governance allows an intermediate tier, it is the cheapest accuracy lever available.

At this scale, federation is a governance choice rather than a bandwidth saving. Eight months of half-hourly training readings from one household occupy about 140 kB, roughly twice a single message of our small LSTM. Thirty rounds of federated training therefore move several times more data than uploading the readings would, and even the traffic needed to reach near-centralised accuracy is of the same order as the data. The case for FL in residential forecasting rests on data minimisation, consent and regulation, not on network load. This agrees with the qualitative argument of Briggs et al. [20] and qualifies the analytic estimates of large savings [18, 19], which depend on model size, sampling rate and the number of rounds.

Two further steps are cheap and worth taking. One epoch of local fine-tuning costs no communication, closed much of the per-household gap between FedAvg and local training, and cut the share of homes harmed by federation from about two thirds to under one half; we would not deploy a federated household forecaster without it. The round should also be chosen on validation error. Validation error rose for a while in the early rounds of

Table 7: Equity of federation relative to local-only training. P90 MAE is the 90th percentile of household test MAE. FT: one epoch of local fine-tuning of the final global model.

Arm	Harmed vs local (%)	Median benefit (kWh)	P90 MAE	Harmed after FT (%)
Centralised	28	+0.0016	0.1364	–
FedAvg, random	56	-0.0005	0.1439	46
FedAvg, α=0.3	66	-0.0013	0.1445	48
FedAvg, cluster	61	-0.0008	0.1428	46
HA-FedProx, α=0.3	66	-0.0013	0.1454	55
FedAvg, K=100	82	-0.0027	0.1514	61
HA-FedProx, K=100	85	-0.0028	0.1515	67

every federated configuration, so a short fixed budget can deliver a worse model than a longer one with validation-based selection.

Finally, operators should look at the high-consumption tail. The homes that gain most from pooled data are the largest consumers, and they are also the ones FedAvg serves worst. A distribution system operator using federated forecasts for congestion management should therefore evaluate errors per consumption tier and not only on average.

6.3. On adaptive proximal coefficients

HA-FedProx was designed on the premise that the dispersion of client updates measures heterogeneity. The premise is half right. The dispersion ratio does rank partitions by heterogeneity, but within a run it is dominated by the convergence of the global model: as the shared descent direction shrinks, any residual disagreement, including pure stochastic-gradient noise, makes the ratio approach one. A schedule proportional to ρ_t therefore increases the proximal pull late in training regardless of the data. Two corrections suggest themselves: normalising ρ_t by its value under a random partition, which requires a reference the server does not have, or using only early-round dispersion, which our data show is the most heterogeneity-sensitive. We leave both to future work. Our main recommendation is a caution: dissimilarity estimates computed from client updates (here, an update-space version of FedProx's B-dissimilarity) confound heterogeneity with the stage of optimisation, and any rule that adapts regularisation to them should be checked against this confound.

6.4. Limitations

The study covers one dataset, one city, one year and one horizon: a winter test period and 30-minute-ahead forecasts for 100 standard-tariff London homes. Its conclusions concern a small LSTM. Larger models raise per-message traffic in proportion, which would strengthen the communication finding but might change the accuracy results.

The network and the stragglers are simulated, and traffic counts payload bytes only (protocol overhead was measured at about 1 %). Keeping data local is not a formal privacy guarantee: updates can leak information [63, 64], and we applied neither differential privacy nor secure aggregation. Both would add accuracy or communication costs on top of those reported here.

The learning rate, model size and input window were fixed in advance and shared by all arms; only the proximal coefficients were tuned. A per-arm search could move individual numbers, although the controlled design should make such shifts affect all arms alike. Lastly, for efficiency the simulator scores the global model on every household's validation and test windows centrally. In a deployment the clients would compute these scores and report only the metrics.

7. Conclusion

We studied federated LSTM forecasting of household load under a protocol that holds the data, model and budget fixed and changes one factor at a time, on 100 London homes from the Low Carbon London trial.

Three conclusions follow. First, the federated accuracy gap is a fragmentation cost: the federated protocol on a single client matches centralised training, and the error grows with the number of clients the same homes are spread over, to about 10 % when every home is its own client. Second, heterogeneity is mainly a communication cost. Regrouping homes from random to fully profile-clustered barely moved final accuracy but raised the bytes needed to approach centralised accuracy by about 85 %. Third, FL is not the bandwidth-light option here. With a small LSTM and half-hourly data, federated training moves more bytes than the readings it protects, so its justification is governance.

For practitioners, this suggests federating a few aggregated clients rather than individual meters, fine-tuning the global model locally (which removed most of the harm to individual households at community scale), and selecting the round on validation error. Proximal regularisation brought no benefit. A

proximal coefficient scheduled from update dissimilarity was worse, because that signal is dominated by the stage of convergence. This caution applies to any regulariser adapted to client-update statistics.

Future work should test these conclusions on other datasets, climates and horizons, and with larger models, where per-message cost grows. It should also add differential privacy and secure aggregation, whose costs would compound those measured here, and design heterogeneity signals that are invariant to the stage of training.

Data and code availability

The Low Carbon London smart-meter data are publicly available from the London Datastore (`https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d`) and are not redistributed by the authors. All code, configuration files, per-run results (summaries, per-round histories and per-household errors) and the scripts that generate every table and figure in this paper are available at [repository URL to be inserted on acceptance; provided to reviewers as a supplementary archive].

CRediT authorship contribution statement

Pushkal Gupta: Conceptualization, Methodology, Software, Data curation, Investigation, Formal analysis, Visualization, Writing – original draft. **Vanka Kaushik:** Conceptualization, Investigation, Validation, Writing – review & editing. **Kalaavathi B:** Supervision, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to help write and test the software pipeline and to draft and edit the text of the manuscript. After using this tool, the authors reviewed and edited the code, the results and the content as needed and take full responsibility for the content of the published article.

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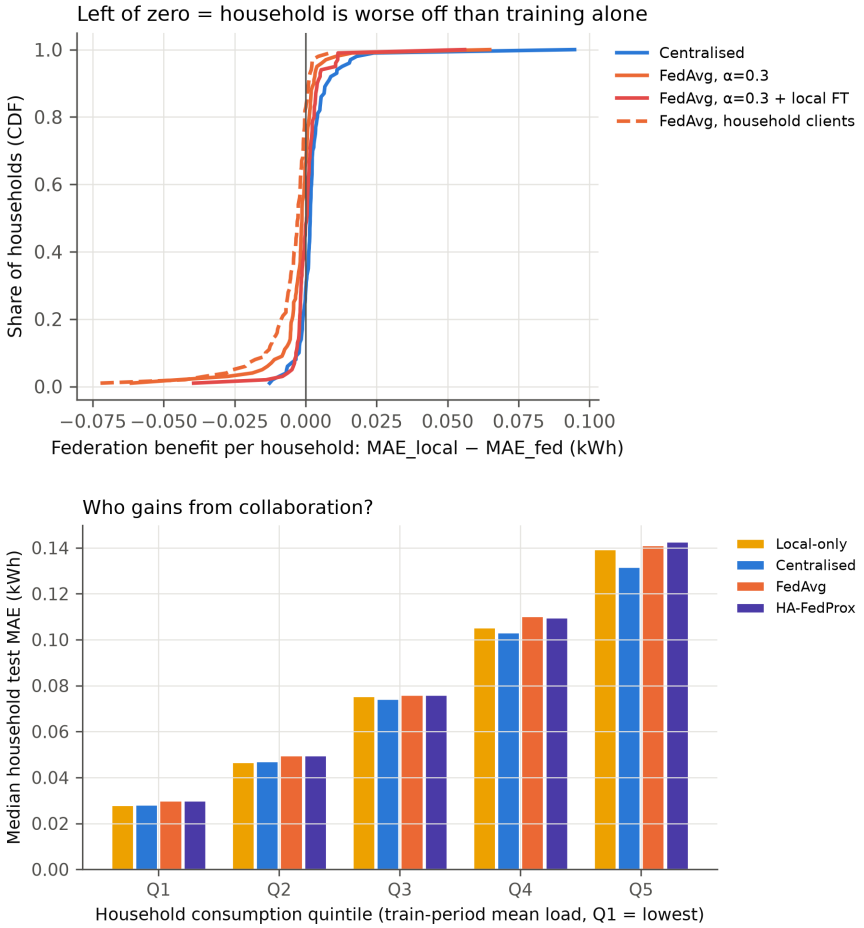


Figure 6: (a) Distribution over households of the benefit of each collaborative arm over local training ($\mathrm{MAE}_{\mathrm{local}} - \mathrm{MAE}_{\mathrm{arm}}$; left of zero means worse off). (b) Median household test MAE by quintile of training-period mean load.